

Uncovering fMRI task structure from brain-state dynamics using Hidden Markov Models

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INTRODUCTION

General Linear Model (GLM) has been the most widely used method for analysis of task fMRI for identifying localized task-evoked activation. It assumes that brain activity is strictly time-locked to predefined task events and thus is not suited to capture time varying neural dynamics crucial in complex task designs. Hidden Markov Models (HMMs) have recently gained popularity to characterize these dynamics as a sequence of probabilistic transitions between a finite set of latent states, where each state has its own connectivity pattern. HMM only uses statistical properties of data, rather than using task information as regressors. This provides a strong basis for validation: if the state activations obtained from HMMs align with the task structure, it suggests the model has captured the neural dynamics accurately. Recent literature has showed that even though HMM is an unsupervised generative model, the state transitions are highly predictive of task load^{1,2}. HMMs have been shown to autonomously segment the data into encoding and retrieval phases in a working memory task. However, these methods assume that states are mutually exclusive and a single state encodes a specific task structure. There is a lack of research that focus on combining the state probabilities from HMM to reconstruct a complex event related task structure in fMRI. In this study, we propose that a complex task dynamic in a pattern separation task can be reconstructed using a weighted linear combination of the state activations. Geometric constraints are applied to the weights to ensure model remains interpretable and within bounds.

METHODS

A total of 87 participants were enrolled in the study, distributed across 2 groups, specifically, 46 with normal cognition (NC) and 41 patients with mild cognitive impairment (MCI). MRI data were acquired on a 7T Siemens Magnetom Terra scanner using a 32-channel head coil. Each fMRI session was collected with the following parameters: 1.5mm isotropic voxel size, 81 slices, TR = 1530ms, TE =

22ms, flip angle = 70 degrees, echo spacing = 0.67 ms and 510 timepoints. A total of 3 task fMRI sessions were collected from each participant. In addition, T1 MP2-RAGE and GRE field mapping scans were also collected.

The pattern separation task used in the study consisted of three phases, namely, encoding, recognition and distraction. In this study, we primarily focus on the recognition phase, during which stimuli presented are (a) same as the ones presented during encoding (22/86), (b) similar to encoding stimuli but distinct (44/86) or (c) completely new (20/86). The subjects responded if the objects seen were same (target), similar (lures) or new (foil). Complete details of the task are discussed in³.

fMRI data were preprocessed as follows: slice-timing correction, realignment and distortion correction, Gaussian smoothing with 4mm FWHM, co-registration to the T1 space and spatially normalization to MNI 2mm space. The preprocessed data had dimensions 510 x 91 x 109 x 91. Data only from brain voxels were extracted resulting in a 510 x 228453 feature dimension. A group Independent Component Analysis (ICA) based on FastICA⁴ with tanh nonlinearity was performed and 2 components representing DMN and FPN were shortlisted for subsequent analysis. Subject-level data were then back-reconstructed using GIG-ICA⁵. Only timepoints of the recognition phase ($T = 292$) were used. Parameters were estimated using an Expectation-Maximization (E-M) algorithm which was run for 20 random realizations and the run with maximum log-likelihood was used. A multisequence model was run separately for both the groups to estimate both HMM parameters (A, π) and GMM parameters (w_i, μ_i, Σ_i). A so-called Hungarian algorithm was run for 100 permutations to match the hidden states across groups.

A separate second-level HMM was then trained using the estimated parameters to obtain posterior state probabilities of hidden states for each session. Mean posterior probabilities were then estimated for each group. Since the task timings were consistent across all sessions and subjects, the canonical hemodynamic response function (HRF) is a good representation of the dynamic task structure. In the pattern separation task, 4 HRFs were of interest – distraction, target, lures and foil. To estimate which hidden states corresponded to the HRFs, a linear optimization that maximized Pearson correlation was utilized. HRF was considered a target timeseries ($1 \times T$) and posterior probabilities were used as the data matrix ($m \times T$). Model was run for each HRF separately to estimate the coefficients for each hidden state. A L2 norm constraint is applied on the coefficients to ensure the weights are comparable and prevents them from growing infinitely large. However, there is a risk that the optimized weights may not reflect actual physiological coupling but rather the spurious correlation present inherently in the data. So, a permutation testing approach was developed to create a null distribution and estimate correlation weights representative of true relationship. Each HRF was circular shifted by a random parameter for 1000 permutations and

null weights were obtained. Circular data shifting ensures the autocorrelation structure of fMRI data is preserved while breaking the temporal alignment. Once the null weights were obtained, the p-values were estimated based on how often the magnitude of the null weights exceeds the true correlation weights. A threshold of $p < 0.05$ was used to test significance.

METHODS FIGURE (OPTIONAL)

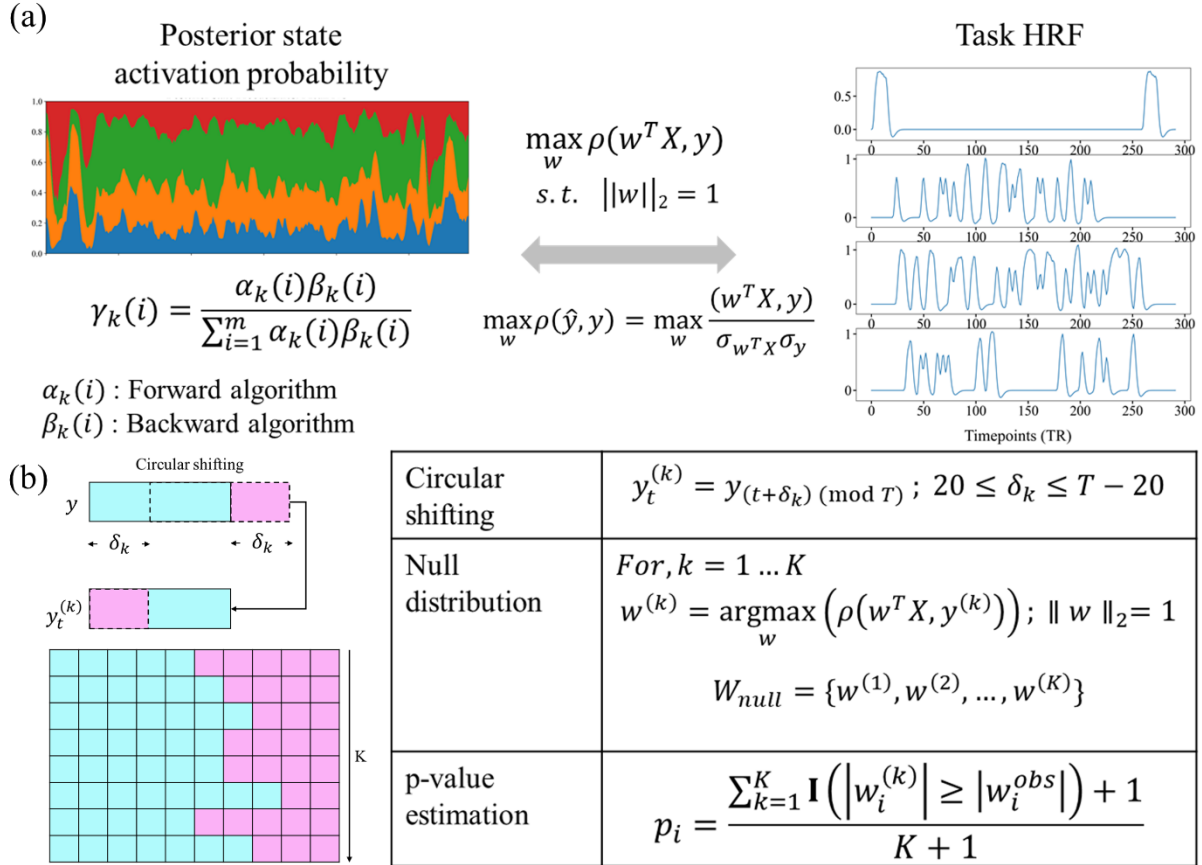


Figure 1: (a) shows the maximization of the Pearson correlation between the state posterior probabilities and task HRF subject to a L2 norm constraint on the coefficients, (b) illustrates the use of circular shifting to generate a null distribution of weights and use it for permutation testing of weights, w obtained from (a).

RESULTS

The homogenous multisequence E-M algorithm used reliably converged across different random seeds. Six was chosen as the optimal number of hidden states as identified by AIC and BIC. With respect to the weights, w obtained during correlation maximization, each state showed varying levels and nature of relationship to the task HRF. For instance, only coefficients of state 1 were found to be positive for the reconstructed distraction HRF for both NC and MCI (w_1 for NC = 0.87 and w_1 for MCI = 0.87). Reconstructed HRF using all states revealed very high correlation of $r_{NC} = 0.8$ and $r_{MCI} = 0.91$. This high r value may primarily be

due to the distinct peaks in the HRF compared to the others. The same state was also identified to be significantly different based on permutation testing ($p=0.001$ for NC and $p=0.04$ for MCI). State 2, however, showed the highest positive coefficients (w_2 for NC = 0.63 and w_2 for MCI = 0.67). Circular shifting also revealed a similar story with $p=0.08$ for NC and $p=0.001$ for MCI). The reconstructed HRFs showed a reduced correlation compared to distraction ($r_{NC} = 0.66$ and $r_{MCI} = 0.5$). Foil condition, representing new stimuli condition, was modulated by state 6 (w_6 for NC = 0.71 and w_6 for MCI = 0.69) and permutation testing results ($p=0.49$ for NC and 0.02 for MCI). Correlation of the reconstructed HRF with true HRF was greater than 0.5 as observed for the previous two HRFs. The target condition HRF showed the weakest correlation with the true value ($r_{NC} = 0.38$ and $r_{MCI} = 0.39$). Accordingly, no significant relationship was identified from the permutation testing highlighting the correlation structure observed is primarily due to noise. Interestingly, only the MCI group showed a negative coefficient for state 6 (w_6 for MCI = -0.31 and w_6 for MCI = -0.31) for both lure and target conditions. While the reconstructed HRFs from MCI showed higher correlation compared to NC for the distraction condition, other conditions showed decreased or comparable correlation in MCI compared to NC. This may possibly reflect the underlying cognitive processing deficits present in MCI.

RESULTS FIGURE (OPTIONAL)

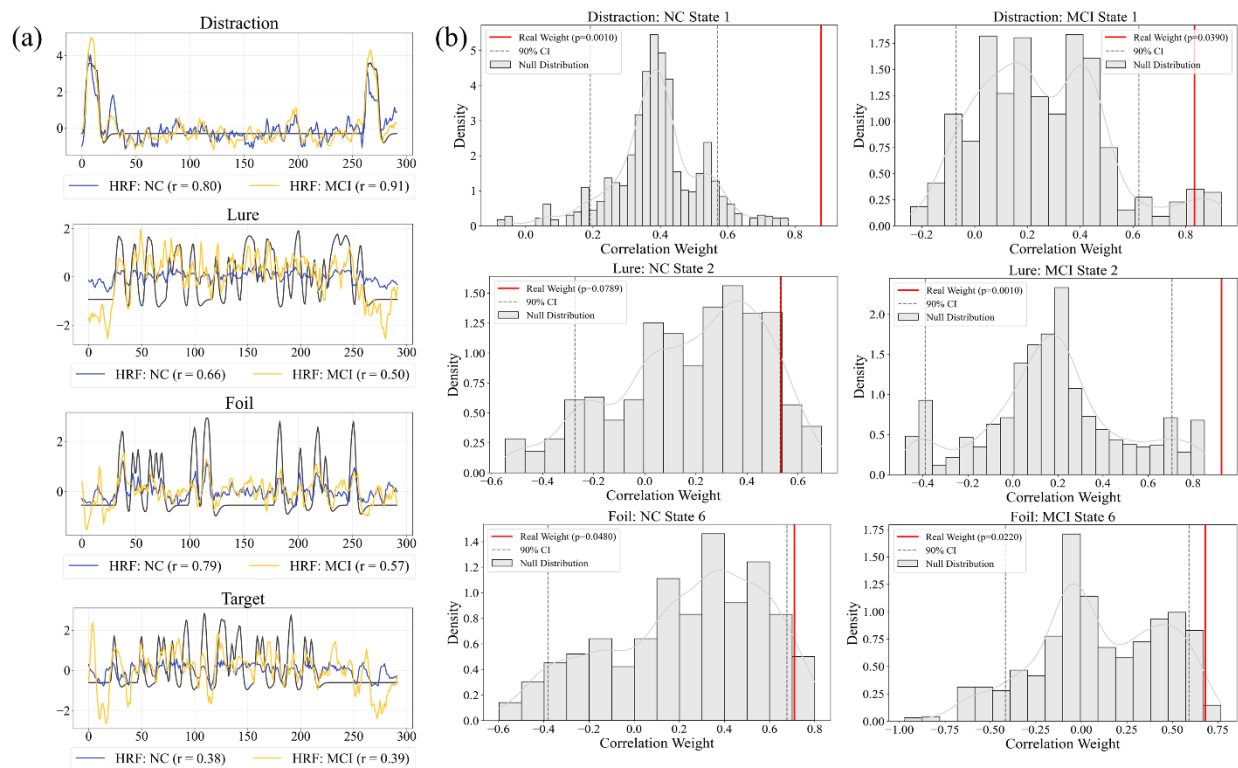


Figure 2: (a) shows results of maximization of correlation between the true HRF and the HRF reconstructed from HMM state probabilities, (b) shows the results of permutation testing highlighting distinct states related to specific task HRFs.

CONCLUSION

We have shown that posterior state probabilities from HMMs can be utilized to reconstruct task HRFs in an unsupervised manner with reliable accuracy in an event related paradigm. Correlation maximization approach paired with permutation testing revealed that distinct brain states are related to specific task events, however, they are collectively needed to describe the entire task structure. However, it is worth noting that only training performance is currently presented and the model's performance needs to be validated using cross-validation, leave-one-out validation or similar methods. This analysis will be further expanded in the future to evaluate the non-linear relationships to overcome the current linear approximation. The potential of these state probabilities in predicting the disease status will also be explored.

REFERENCES/CITATIONS

1. Quinn AJ, Vidaurre D, Abeyesuriya R, Becker R, Nobre AC, Woolrich MW. Task-Evoked dynamic network Analysis through hidden Markov modeling. *Frontiers in Neuroscience*. 2018;12:603. doi:10.3389/fnins.2018.00603
2. Liu L, Jiang J, Li H, Ding G. Tripartite organization of brain state dynamics underlying spoken narrative comprehension. *eLife*. January 2025. doi:10.7554/elife.99997.2
3. Yang Z, Zhuang X, Koenig KA, et al. Pattern separation involves regions beyond the hippocampus in non-demented elderly individuals: A 7T object lure task fMRI study. *Imaging Neuroscience*. 2024;2:1-15. doi:10.1162/imag_a_00404
4. Hyvarinen A. Fast and robust fixed-point algorithms for independent component analysis. *IEEE Transactions on Neural Networks*. 1999;10(3):626-634. doi:10.1109/72.761722
5. Du Y, Fan Y. Group information guided ICA for fMRI data analysis. *NeuroImage*. 2012;69:157-197. doi:10.1016/j.neuroimage.2012.11.008

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